EX No:04

Date:

Linear Regression Model Building using Boston Dataset

4. This problem involves the Boston data set, which we saw in the lab for this chapter. We will now try to predict per capita crime rate using the other variables in this data set. In other words, per capita crime rate is the response, and the other variables are the predictors.

LOAD LIBRARY:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
from ISLP import load_data
from ISLP.models import (ModelSpec as MS, summarize, poly)
```

LOAD DATA:

```
df = load_data("Boston")
```

(a) For each predictor, fit a simple linear regression model to predict the response. Describe your results. In which of the models is there a statistically significant association between the predictor and the response? Create some plots to back up your assertions.

CODE:

```
y = df["crim"]

predictors = [col for col in df.columns if col != "crim"]
for predictor in predictors:

    print("Simple Linear Regression Model")
    print("CRIM ~", predictor)
    X = sm.add_constant (df[[predictor]])
    model = sm.OLS(y, X).fit()
    print(model.summary())
```

OUTPUT:

```
Simple Linear Regression Model
CRIM ~ zn
                            OLS Regression Results
==============================================================================
Dep. Variable:                   crim   R-squared:                       0.040
Model:                            OLS   Adj. R-squared:                  0.038
Method:                 Least Squares   F-statistic:                     21.10
Date:                Tue, 13 Oct 2026   Prob (F-statistic):           5.51e-06
Time:                        10:29:59   Log-Likelihood:                -1790.0
No. Observations:                 506   AIC:                             3596.
Df Residuals:                     504   BIC:                             3604.
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          4.4537      0.417     10.675      0.000       3.634       5.273
zn            -0.0739      0.016     -4.594      0.000      -0.106      -0.042
==============================================================================
Omnibus:                      567.443   Durbin-Watson:                   0.857
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            32753.004
Skew:                           5.257   Prob(JB):                         0.00
Kurtosis:                      40.986   Cond. No.                         28.8
==============================================================================
...
------------------------------------------------------------------------------
```

PLOTS:

```
sns.pairplot( df, y_vars="crim",  x_vars=[col for col in df.columns if col != "crim"],
                kind="reg", height=3 )

plt.show()

sns.pairplot(df, diag_kind="kde")

plt.show()

# Calculate correlation matrix
correlation_matrix = df.corr()

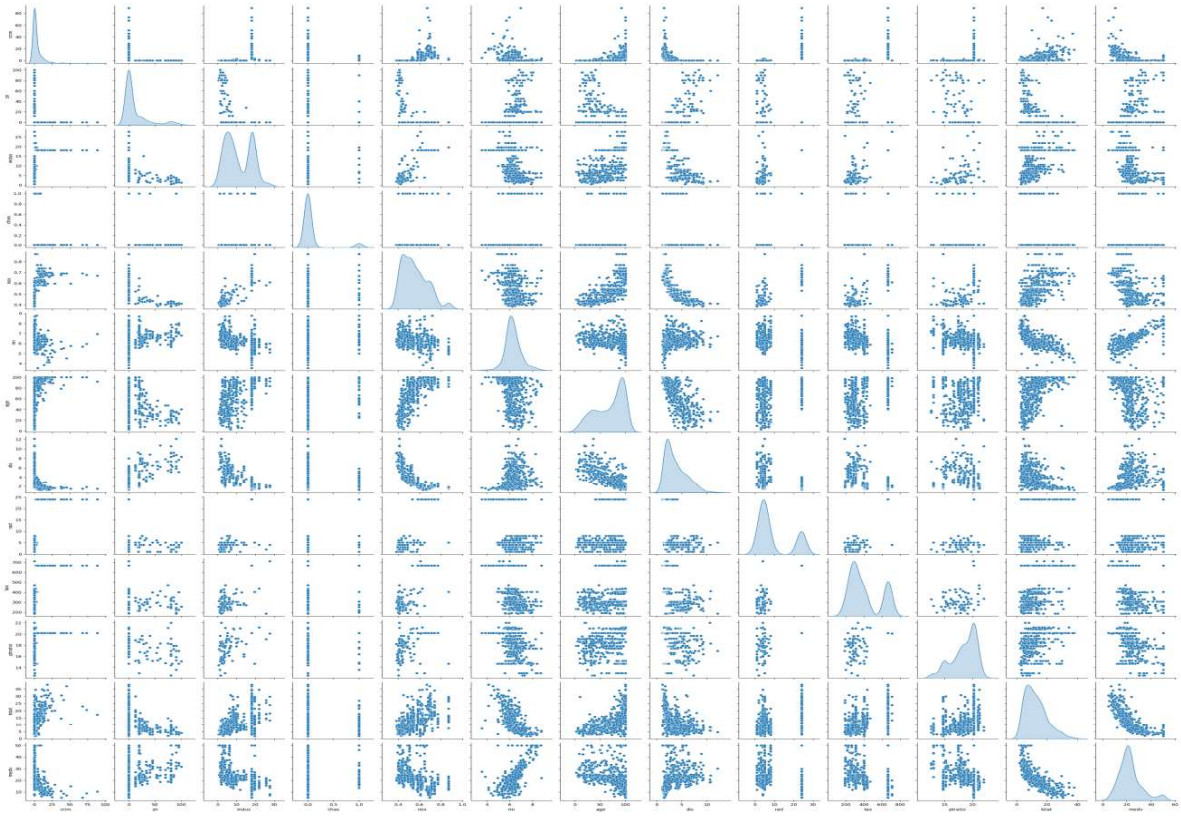
# Create heatmap
plt.figure(figsize=(8, 8))

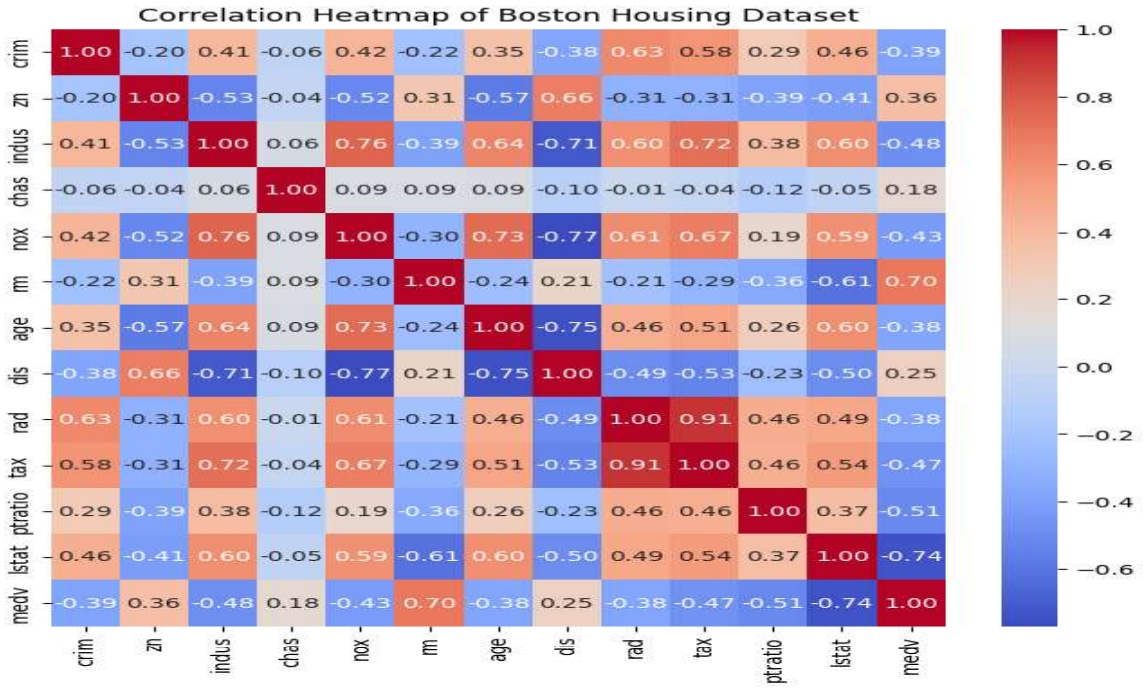
sns.heatmap(correlation_matrix,
                annot=True,
                fmt=".2f",
                cmap="coolwarm")

plt.title("Correlation Heatmap of Boston Housing Dataset")

plt.show()
```

OUTPUT:





(b) Fit a multiple regression model to predict the response using all of the predictors. Describe your results. For which predictors can we reject the null hypothesis H0 : βj = 0 ?

CODE:

```
X = sm.add_constant(df.drop("crim", axis=1))
y = df["crim"]
multiple_model = sm.OLS(y, X). fit ()
print(multiple_model.summary())
```

OUTPUT:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                   crim   R-squared:                       0.449
Model:                            OLS   Adj. R-squared:                  0.436
Method:                 Least Squares   F-statistic:                     33.52
Date:                Wed, 26 Aug 2026   Prob (F-statistic):           2.03e-56
Time:                        09:51:06   Log-Likelihood:                -1655.4
No. Observations:                 506   AIC:                             3337.
Df Residuals:                     493   BIC:                             3392.
Df Model:                          12
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         13.7784      7.082      1.946      0.052      -0.136      27.693
zn             0.0457      0.019      2.433      0.015       0.009       0.083
indus         -0.0584      0.084     -0.698      0.486      -0.223       0.106
chas          -0.8254      1.183     -0.697      0.486      -3.150       1.500
nox           -9.9576      5.290     -1.882      0.060     -20.351       0.436
rm             0.6289      0.607      1.036      0.301      -0.564       1.822
age           -0.0008      0.018     -0.047      0.962      -0.036       0.034
dis           -1.0122      0.282     -3.584      0.000      -1.567      -0.457
rad            0.6125      0.088      6.997      0.000       0.440       0.784
tax           -0.0038      0.005     -0.730      0.466      -0.014       0.006
ptratio       -0.3041      0.186     -1.632      0.103      -0.670       0.062
...
```

INTERPRETATION:

- (a) Each simple regression tests one predictor at a time. For zn the slope is -0.0739 ($p < 0.001$), so it is statistically significant, but R-squared is only 0.040: it explains just 4% of the variation in crime rate. The output shown is truncated after the first model (zn).
- The plots and heatmap show crim is positively related to rad (0.63), tax (0.58), lstat (0.46), nox (0.42), indus (0.41) and age (0.35), and negatively to medv (-0.39), dis (-0.38), rm (-0.22) and zn (-0.20). The chas correlation is only -0.06, too small to be significant with 506 tracts, so chas is the one predictor with no clear association.
- crim is extremely right-skewed (skew 5.26, kurtosis 41 in the zn model): most tracts have near-zero crime and a few have very high crime, so residuals are far from normal. A log transformation of crim would be worth trying.
- The predictors are collinear: rad and tax correlate at 0.91, indus-nox 0.76, nox-dis -0.77 and age-dis -0.75, which complicates interpreting individual coefficients in the multiple model.
- (b) The multiple regression has R-squared 0.449 (adjusted 0.436) and F = 33.52 (p of about 2e-56), so together the predictors explain about 45% of the variation in crime rate and the overall model is highly significant.
- Among the coefficients visible in the output, we can reject H0: $\beta_j = 0$ for zn (0.0457, $p = 0.015$), dis (-1.0122, $p < 0.001$) and rad (0.6125, $p < 0.001$); nox is borderline ($p = 0.060$) and indus, chas, rm, age, tax and ptratio are not significant. The printed output cuts off after ptratio, so lstat and medv are not shown.